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Multimodal probing of oxygen and water interaction with metallic and semiconducting carbon nanotube networks under ultraviolet irradiation

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Abstract. Interaction between ultraviolet (UV) light and carbon nanotube (CNT) networks plays a central role in gas adsorption, sensor sensitivity, and stability of CNT-based electronic devices. To determine the effect of UV light on sorption kinetics and resistive gas/vapor response of different CNT networks, films of semiconducting single-wall nanotubes (s-SWNTs), metallic single-wall nanotubes, and multiwall nanotubes were exposed to O_2 and H_2O vapor in the dark and under UV irradiation. Changes in film resistance and mass were measured *in situ*. In the dark, resistance of metallic nanotube networks increases in the presence of O_2 and H_2O, whereas resistance of s-SWNT networks decreases. UV irradiation decreases the resistance of metallic nanotube networks in the presence of O_2 and H_2O and increases the gas/vapor sensitivity of s-SWNT networks by nearly a factor of 2 compared to metallic nanotube networks. s-SWNT networks show evidence of delamination from the gold-plated quartz crystal microbalance crystal, possibly due to preferential adsorption of O_2 and H_2O on gold. UV irradiation increases the sensitivity of all CNT networks to O_2 and H_2O by an order of magnitude, which demonstrates the importance of UV light for enhancing response and lowering detection limits in CNT-based gas/vapor sensors. © *2016 Society of Photo-Optical Instrumentation Engineers (SPIE)* [DOI: 10.1117/1.JPE.6.025506]

Keywords: carbon; nanotubes; ultraviolet; water; oxygen; sorption; quartz crystal microbalance.

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1 Introduction

Carbon nanotubes (CNTs) belong to a family of carbon allotropes that are being explored for application in transparent conducting electrodes,[1,2] photovoltaics,[3] light-emitting diodes,[4] transistors,[5] biointerfaces,[6] and gas sensors.[7] Their unique optoelectronic properties and compatibility with roll-to-roll processing have brought carbon-based materials to the forefront of nanoscience research.[8] The bandgap of semiconducting CNTs can be further tuned with intrinsic and extrinsic dopants.[9] However, the stability of CNT devices in humid- and oxygen-rich environments may present a fundamental limitation to reliable device performance. The high surface area of CNTs facilitates adsorption of gas molecules such as water and oxygen, which can alter bandgap and work function, local density of states, and induce charge transfer and semiconductor-metal transitions.[10–12]

The perfect sp^2 hybridization in defect-free graphene makes it electronically stable and generally prevents strong chemical adsorption of atmospheric gases.[11] The curvature of fullerenes and

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nanotubes on the other hand introduces sp^3 character to the dominant sp^2 planar C═C bonding.[8] Disruption of the perfect sp^2 network significantly increases the reactivity of nanostructures. Exposure of CNT networks to oxygen and humidity leads to charge transfer with O_2 and H_2O adsorbates.[10–15] Rapid desorption of O_2 and H_2O from film surfaces is critical for the reversibility of CNT sensors and strongly affects the surface wetting characteristics of CNT networks.[10–12] Thermal treatment is often used for the removal of physisorbed molecules, but in CNTs it may promote material decomposition, resulting in the formation of CO and CO_2 products rather than O_2 desorption.[16] Partial CNT oxidation disrupts sp^2 bonding and leads to the formation of locally electron-rich and electron-deficient regions/defects, facilitating the formation of COOH, OH, and CO functional groups.[11,17] In addition to thermal treatment, ultraviolet (UV) irradiation can be used for desorption of adsorbates and control of wetting properties.[11,15] Continuous UV irradiation of CNTs leads to the formation of structural defects in networks,[15] electronic photoexcitations in gas molecules and networks,[16] and photoinduced plasmons in the CNTs.[18,19] This combination of processes makes it difficult to identify the primary mechanisms that mediate adsorption/desorption processes under UV irradiation, which generally improves the resistive response of CNTs.[17]

Density function theory calculations show that physisorbed O_2 molecules orient with O═O bonds parallel to CNT surfaces.[15] This leads to hybridization between the valence bands of CNTs and O_2 acceptor molecules, which increases the density of states at the Fermi level[14] and results in hole doping behavior.[16] Oxygen molecules physisorb strongly (with binding energy of ~0.25 eV) on SWNTs, withdrawing about e/10 charge per each adsorbed molecule.[20] Similar response to O_2 was reported for multiwall nanotube (MWNT),[17] C_{60},[21–23] and other graphene-derived structures.[18,20,24] Desorption of O_2 under UV irradiation is primarily mediated by photogenerated π-electron surface plasmons.[17] Plasmon lifetime in SWNTs is on the order of 1 fs, after which plasmons decay into single-particle hot electron and hole excitations.[25] Photoinduced plasmon energy of hot carriers can overcome the activation energy (E_A) barrier of O_2 adsorption.[18,20] Hot carrier injection has been proposed as one of the primary mechanisms of UV photodesorption of O_2 and other electron-accepting adsorbates from SWNT, MWNT, and grapheme.[17,19]

UV irradiation also enhances dissociative chemisorption of O_2 and H_2O on carbon surfaces. Singlet oxygen can be formed by UV photoexcitation of gas-phase triplet O_2 or by interactions between physisorbed triplet O_2 and UV photoexcited carbon allotropes.[17] Exposure to singlet O_2 generally leads to the oxidation of CNT networks.[22,26] During dissociative chemisorption of oxygen, each O atom rests above a C═C bond bridge site to form a C—O—C complex, which destroys the perfect π-bonding between carbon atoms. It remains unclear whether this enhances further photo-oxidation at other C—C bridge sites[15,16] due to disruption of π-bonding or leads to chemical stabilization of the nanostructure by trapping local photoinduced charge carriers.[17,22]

Dissociative chemisorption of H_2O becomes favored over O_2 when oxygen or hydrogen defects are present on the CNT surface. UV-enhanced removal or addition of C—H and C—OH groups formed by H_2O dissociation allows for control of reversible hydrophobic–hydrophilic transitions on the surface,[15] where the OH moiety acts as an electron acceptor, behaving similarly to physisorbed O_2.[11] Charge transfer to H_2O and OH has been observed in SWNT[14] and C_{60}[21] and can lead to bandgap opening in graphene by up to 0.2 eV.[12,15]

Photoinduced adsorption/desorption processes can be described in terms of electronic excitations in CNTs in the presence of electron acceptors such as O_2 or H_2O, as shown in Fig. 1(d). E_{A1} represents the energy barrier to charge transfer from the CNT to an electron acceptor (O_2/H_2O), and E_{A2} is the energy barrier to electron transfer from the acceptor to the CNT. O_2 approaches the CNT in its spin triplet ground state, where the O—O bond has a binding energy of 5.2 eV, the highest occupied molecular orbital is the π^*_{2p} level and the lowest unoccupied molecular orbital (LUMO) is the σ^*_{2p} level.[27] In the dark, E_{A1} corresponds to the work function of the carbon surface (~4.8 eV for defect-free SWNT bundles[28] and 4.7 eV for defect-free C_{60})[29] and adsorption of O_2 is primarily mediated by van der Waals interactions. For purely van der Waals-induced adsorption processes, O_2 remains in its triplet ground state and $E_{A1} = 0$. However, the presence of structural defects enables charge transfer between O_2 and SWNT with $E_{A1} \approx 1.2$ eV.[17,30] Physical adsorption of O_2 results in pinning and broadening of the O_2 LUMO level to the Fermi level (E_F) of the CNT, effectively decreasing the energy barrier E_{A1} and increasing the probability of charge transfer complex ($CNT^{\delta+\cdots}O_2^{\delta-}$) formation.[31] The formation of this unstable intermediate state is denoted by (1), while desorption is denoted by (2).

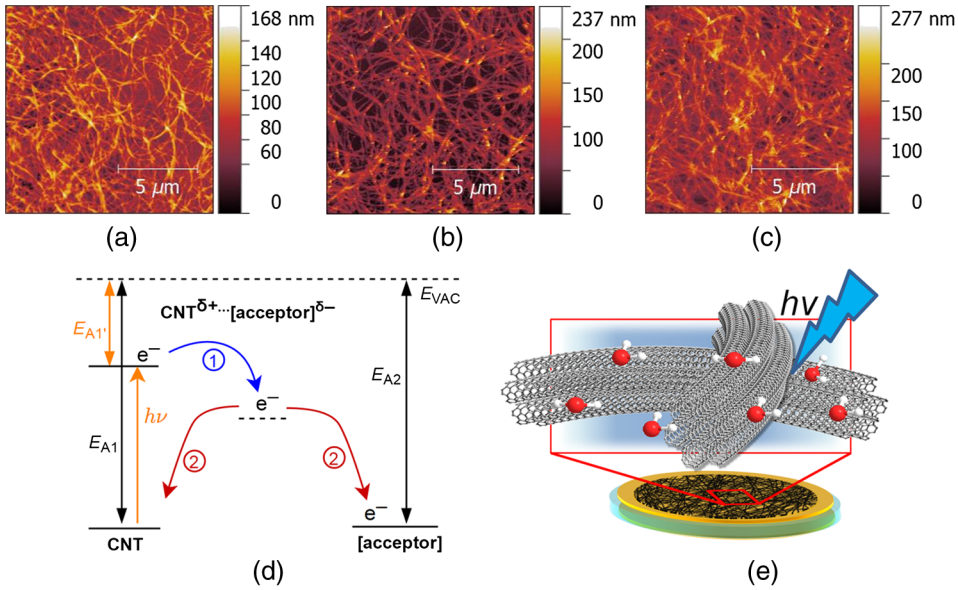

Fig. 1 AFM images of (a) s-SWNT, (b) m-SWNT, and (c) MWNT networks. (d) Schematic of photoinduced adsorption and desorption of electron-accepting O_2 and H_2O on CNTs. The energy barrier to electron transfer to (E_{A1}) and from (E_{A2}) the charge transfer complex (CNT$^{\delta+\cdots}$[acceptor]$^{\delta-}$) is decreased under UV irradiation, resulting in enhancement of both (1) photoinduced chemisorption and (2) photoinduced molecular desorption. (c) Schematic of H_2O adsorption and photoinduced desorption from a junction between two bundles of (5, 5) SWNT bundles on a gold-plated QCM crystal.

Under UV irradiation, photon absorption in CNTs may promote E_F level electrons by as much as 3.5 eV, leading to electron–hole pair production and plasmon generation, which facilitate electron transfer from the CNT to physisorbed O_2 by reducing E_{A1} to $E_{A1'}$. In semiconducting carbon structures, adsorption-induced bending of the valence and conduction bands causes charge carriers to migrate in opposite directions. In p-type CNTs, UV absorption enhances downward band bending since the surface photovoltage (SPV) transports negative charge to the film surface.[32] The SPV effect narrows the effective width of the E_A barrier, increasing the probability of charge transport back and forth across the barrier. Excitation in the adsorbed O_2 molecule by singlet quenching of the CNT surface or by direct absorption of UV photons[17] lowers the effective height of E_{A2}, which increases the probability of electron transfer from adsorbed O_2 back to the CNT, facilitating molecular desorption. Since UV irradiation effectively lowers energy barriers E_{A1} and E_{A2}, both molecular adsorption and desorption rates may be enhanced in the presence of UV light.

Figures 1(a)–1(c) show atomic force microscope (AFM) images of the semiconducting single-wall nanotube (s-SWNT), metallic single-wall nanotube (m-SWNT), and MWNT networks, respectively. The CNTs exist in ∼100-nm diameter bundles over 10-μm long in uniformly dispersed networks. Some impurities are visible in the m-SWNT and MWNT networks. Figure 1(e) shows a schematic of H_2O adsorption on a junction between two bundles of (5, 5) SWNTs. The strong van der Waals interaction between nanotubes is perturbed upon gas/vapor adsorption, allowing gas migration between bundles, which may affect CNT-quartz crystal microbalance (QCM) adhesion at the QCM interface. Adsorption of charge-accepting molecules alters the conductivity of the tubes as well as the tube–tube junctions, which results in the strong electronic gas/vapor response of SWNTs and multiwall nanotubes (MWNTs).

2 Experiment

We studied the effect of UV irradiation on the electrical resistance of s-SWNT, m-SWNT, and MWNT networks in humid- and oxygen-rich environments. Adsorption/desorption of O_2 and H_2O on CNT networks was measured using a QCM. Single-wall nanotubes (s-SWNTs) and

m-SWNTs were purchased from Nanointegris. Dispersion of SWNTs was prepared by ultrasonicating them in toluene for 1 h. MWNTs (99% pure) were purchased from Cheap Tubes Inc. and purified by nitric acid reflux. MWNTs were mechanically mixed in dimethylformamide and sonicated for 30 min. The SWNT and MWNT solutions were drop-cast on AT-cut gold-plated QCM crystals and on Si/SiO_2 substrates with gold interdigitated electrodes. Substrates were precleaned by sonicating in acetone for 30 min. Substrates were weighed with 0.1-μg resolution using a UMX2 microbalance (Mettler Toledo) before and after solution casting to determine the mass of each deposited CNT film. Data collected during the experiment (QCM frequency shift and film resistance) were normalized by CNT film mass in order to compare the response between different networks. AFM images of CNT networks were acquired using an Asylum Research Cypher S AFM in tapping mode with conductive Pt/Ir-coated cantilever operated at 73-MHz resonance frequency.

Prior to measurements, networks were placed under vacuum of 10^{-6} Torr for 3 h to desorb contaminants. During the experiment, samples were exposed to 5-, 10-, 15-, and 20-Torr pulses of H_2O vapor and O_2, which were injected into the vacuum chamber using a gas control manifold. Electrical resistance of the networks was measured in a 4-point probe configuration with a Keithley 6430 source-meter. Frequency shift of the QCM crystal was measured using an SRS QCM200 5-MHz QCM system. Gas flow control, electrical, and mass measurements were integrated and controlled using LabVIEW software. The experiment was performed in the dark and under UV irradiation from an Hg lamp filtered through the quartz window of the vacuum chamber. The QCM data were corrected for the response of an empty QCM crystal under the same experimental conditions.

3 Results and Discussion

Figure 2 shows mass change (ΔM) of CNT networks exposed to 5-, 10-, 15-, and 20-Torr pulses of O_2 (a) and H_2O vapor (b) in the dark and under UV radiation. Resistance change (R/R_0) of the CNT networks during the same gas/vapor sequences is shown in Fig. 2(c) for O_2 and

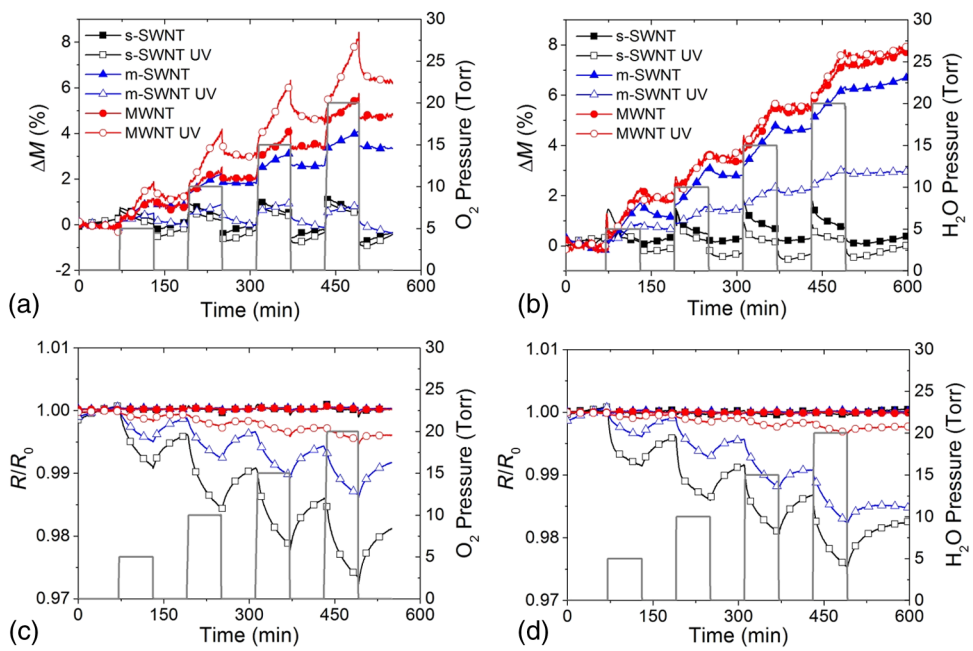

Fig. 2 (a) Mass change (ΔM) of CNT networks when exposed to 5-, 10-, 15-, and 20-Torr pulses of O_2 in the dark and under UV irradiation. (b) ΔM when CNT networks are exposed to 5-, 10-, 15-, and 20-Torr pulses of H_2O in the dark and under UV irradiation. (c) Resistance change (R/R_0) of CNT networks when exposed to pulses of O_2 in the dark and under UV irradiation. (d) Resistance change (R/R_0) of CNT networks when exposed to pulses of H_2O in the dark and under UV irradiation. In (c) and (d), $|R/R_0| < 0.05\%$ for s-SWNT, m-SWNT, and MWNT networks in the dark.

Fig. 2(d) for H_2O. In m-SWNT and MWNT, mass sorption on CNT networks is irreversible upon exposure to vacuum, both in the dark and under UV irradiation. UV irradiation leads to enhancement of the resistive gas/vapor response by roughly an order of magnitude in all CNT networks, although the effect of UV irradiation O_2/H_2O adsorption/desorption kinetics varies across networks.

Since ΔM and R/R_0 response kinetics are largely independent of the pressure of the gas/vapor pulse and the magnitude of response is roughly proportional to the pressure, it is useful to compare the response of each CNT network during the 20-Torr pulse. At the beginning of the 20-Torr pulse, mass and resistance of CNT networks have not fully recovered from the previous gas/vapor pulses. However, ΔM and R/R_0 response kinetics are similar during every gas/vapor pulse, including the 5-Torr pulse, in which CNT networks were fully recovered from previous experiments. This suggests that significant changes in ΔM and R/R_0 response kinetics do not occur as a result of incomplete recovery. Therefore, we describe response kinetics during the 20-Torr pulse in terms of fully recoverable CNT networks. CNT responses are compared during the 20-Torr gas/vapor pulse by subtracting background mass and normalizing resistance so that $\Delta M = 0$ and $R/R_0 = 1$ when the 20-Torr pulse begins. ΔM of the CNT networks exposed to 20-Torr pulses of (a) O_2 and (b) H_2O vapor are shown in Fig. 3. Adsorption of both O_2 and H_2O on s-SWNT, m-SWNT, and MWNT networks leads to rapid initial mass increase in the dark and under UV irradiation.

Upon initial exposure to H_2O, mass of the MWNT network first increases quickly, then decreases, then increases further, but more slowly. The apparent rapid increase and decrease of ΔM within the first 2 min of the pulse are likely a consequence of abrupt pressure change and/or condensation of H_2O on the QCM crystal but are not necessarily related to H_2O adsorption on the MWNT network since the MWNT resistance exhibits a monotonic decrease. The gradual increase in ΔM in m-SWNT and MWNT networks after the initial gas/vapor adsorption step is likely a result of diffusion of O_2 and H_2O into the CNT network. The decrease in ΔM after the initial increase in s-SWNT is discussed further below.

MWNT and m-SWNT networks exhibit similar behavior in the dark. However, UV irradiation increases the rate of oxygen diffusion into the MWNT film by a factor of 2 without significant change in the desorption kinetics. This suggests that UV-induced charge transfer from MWNTs to physisorbed oxygen facilitates chemisorption, which cannot be reversed by exposure to vacuum, even under UV irradiation. Oxygen adsorption on m-SWNT is inhibited under UV irradiation. In m-SWNTs, delocalization of conduction electrons enhances the generation of UV-induced plasmons, which facilitate molecular desorption of O_2 by hot carrier injection.

As shown in Fig. 3(b), desorption of H_2O from m-SWNTs and MWNTs occurs over a time scale of roughly 10 h. Less than half of adsorbed H_2O is desorbed after networks have been placed under vacuum for 1 h, even under UV irradiation. Water molecules that are hydrogen-bonded to C—H and C—OH groups are readily desorbed under vacuum. This makes up <20% of the total H_2O adsorbed. Since UV irradiation does not affect the H_2O adsorption/desorption

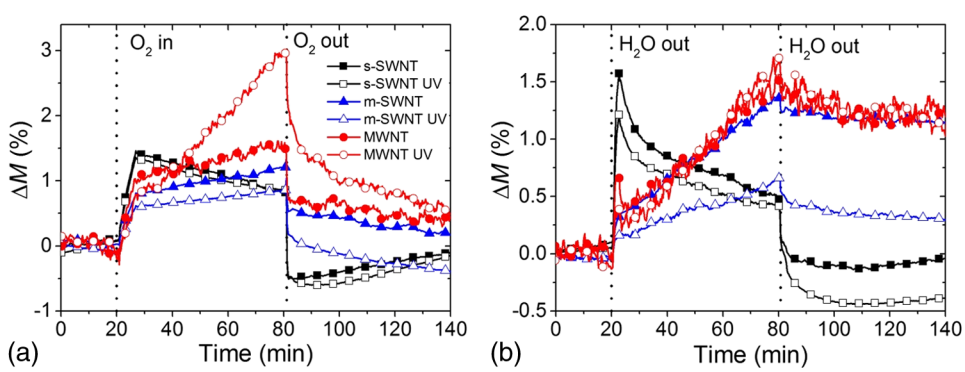

Fig. 3 Change in mass (ΔM) of s-SWNT, m-SWNT, and MWNT networks when networks are exposed to a 20-Torr pulse of (a) O_2 and (b) H_2O in the dark (solid shapes) and under UV irradiation (open shapes). Introduction and removal of gas/vapor are marked by vertical dotted lines.

kinetics on MWNTs, it suggests that the H_2O-MWNT interactions are not strongly affected by photoinduced processes. The irreversible H_2O chemisorption observed on MWNTs occurs primarily through interactions with CNT defects, since the energy barrier to H_2O chemisorption cannot be overcome by charge transfer from defect-free MWNTs.

UV irradiation strongly decreases the rate of H_2O adsorption and diffusion into the m-SWNT network. Irradiation generates surface plasmons, which result in hot carrier injection into O_2 and H_2O adsorbates, stimulating desorption.[25] The time constant for gas desorption is on the order of hours, which suggests that the energy density of photoinduced plasmons in m-SWNTs is below the energy barrier required for facilitating rapid desorption of chemisorbed OH and H groups.

Interestingly, 5 min after the start of the O_2 pulse, mass of the s-SWNT film begins to decrease linearly [Fig. 3(a)]. Over the course of the 60-min O_2 pulse, roughly 30% of the initial mass is lost. A similar decrease in mass is observed 2 min after the H_2O pulse begins, in which 70% of the initially adsorbed H_2O mass is lost [Fig. 3(b)]. Similar behavior was observed for s-SWNT upon exposure to 5-, 10-, and 15-Torr O_2/H_2O pulses. Since the decrease in mass is observed in the dark as well as under UV irradiation, it is not likely caused by photodesorption processes. At first glance, the data suggest that after the initial adsorption step, a different desorption process takes place, while the s-SWNT film is still exposed to gas/vapor during the same pulse. However, monotonically changing film conductivity (see s-SWNT data in Fig. 4) demonstrates that adsorption increases continuously throughout the entire duration of the gas/vapor pulse. We propose that the apparent decrease in mass is evidence of partial delamination of the s-SWNT film from the surface of gold-plated QCM crystal. Since individual SWNTs are bundled primarily as a result of van der Waals interactions, adsorption of O_2 or H_2O on bundled tubes disrupts tube-tube adhesion. As O_2/H_2O molecules diffuse to the film-gold interface, intercalation of adsorbates between s-SWNTs and gold may result in disruption of the tube-gold adhesion and cause partial delamination of s-SWNTs from the QCM crystal due to preferential adsorption of O_2 and H_2O on gold. Delamination facilitates slipping between the s-SWNTs and gold as the crystal oscillates at ~5 MHz. This reduces the effective mass of the film bound to the crystal, increasing the resonance frequency of the crystal. Since the decrease in mass is observed during each gas/vapor pulse, we propose that delamination of the s-SWNT film from the gold surface is reversible and occurs 2 to 5 min after gas/vapor exposure. This description is consistent with the observation that changes in mass become negative ($\Delta M < 0$) after gas/vapor evacuation. Desorption of oxygen from the s-SWNT/QCM interface leads to slow recovery of the gold-nanotube adhesion. The effect of H_2O on s-SWNT delamination is more pronounced than that of O_2 primarily because of hydrogen bonding between H_2O intercalates at the s-SWNT/gold interface.

The change in electrical resistance of CNTs exposed to 20-Torr pulses of (a) O_2 and (b) H_2O in the dark and under UV irradiation are shown in Fig. 4. Prior to the gas/vapor pulse, $R/R_0 < 0$

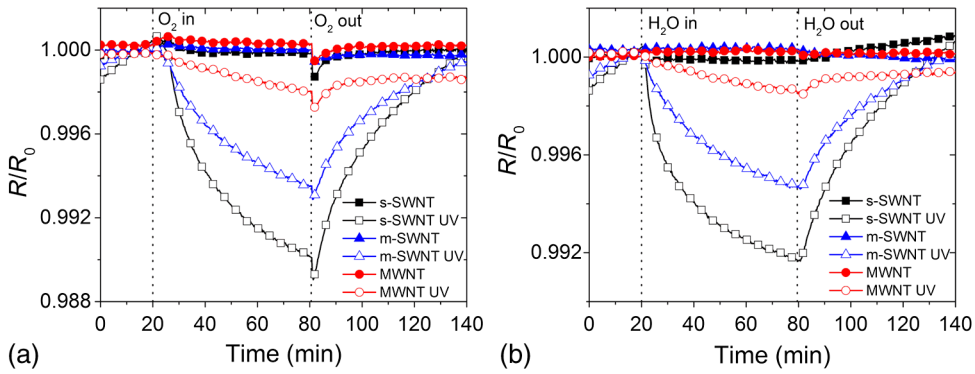

Fig. 4 Resistance change (R/R_0) of semiconducting SWNT (s-SWNT), metallic SWNT (m-SWNT), and MWNT networks when networks are exposed to a 20-Torr pulse of (a) O_2 and (b) H_2O in the dark (solid shapes) and under UV irradiation (open shapes). Introduction and removal of gas/vapor are marked by vertical dotted lines. Prior to the gas/vapor pulse, $R/R_0 < 0$ for metallic CNTs due to lack of full recovery from the previous pulse.

for metallic CNTs due to lack of full recovery from the previous pulse 15-Torr pulse. The time constant of the resistance response is not strongly influenced by the pressure of the gas/vapor. The resistance of s-SWNT, m-SWNT, and MWNT networks changes by <0.05% in response to 20-Torr pulses of O_2 and H_2O in the dark. Under UV irradiation, the magnitude of the gas/vapor response increases by more than a factor of 10. The strong enhancement of gas response by UV irradiation is consistent with results reported in the literature.[33] The weak reversible gas response that occurs when networks are kept in the dark suggests that significant change transfer does not occur between O_2/H_2O and CNTs in the absence of UV irradiation. In MWNTs, UV irradiation increases both the rate of O_2 adsorption and the magnitude of the resistance response. This suggests that enhancement of the resistance response occurs mainly because more O_2 is adsorbed under UV irradiation. However, UV irradiation enhances the MWNT resistance response to H_2O without increasing H_2O adsorption on the film, so it is likely that the resistance response enhancement is related to photogenerated charge carriers, not additional vapor adsorption in the case of H_2O.

The UV-enhanced decrease in resistance in s-SWNTs upon exposure to both O_2 and H_2O confirms that both molecules behave as electron acceptors. Defect-free s-SWNTs lack free charge carriers at E_F[34] but charge transfer from s-SWNTs to adsorbates leaves holes, which contribute to charge transport resulting in a drop of film resistance. In both s-SWNTs and m-SWNTs, the resistance response to O_2 and H_2O is reversible. UV irradiation inhibits reversibility of the MWNT gas/vapor resistive response. This is consistent with the results in Figs. 2 and 3, which show that the O_2/H_2O adsorbed on MWNT during the gas/vapor pulses does not fully desorb under vacuum conditions. It is likely that formation of charge transfer complexes between O_2/H_2O and MWNT result in the irreversible gas binding. In MWNTs, Cao et al.[35] observed chemisorption of reducing gases, which opens a gap between E_F and the valence band, resulting in the formation of a Schottky barrier at the film surface. This explanation is consistent with our results since we observe a resistance drop upon exposure to acceptor adsorbates.

The magnitude of the resistance response (ΔR) of the CNT networks is shown as a function of gas/vapor pressure in Fig. 5(a) for O_2 and Fig. 5(b) for H_2O. For clarity, the horizontal dotted lines denote $\Delta R = 0$. It is clear that the electronic response to O_2 as a function of pressure is qualitatively similar to that of H_2O. An important result in Fig. 5 is that $\Delta R > 0$ in m-SWNTs and MWNTs upon exposure to gas/vapor in the dark, but $\Delta R < 0$ under UV irradiation. The change in the sign of ΔR under UV irradiation suggests that the dominant mechanism of the resistive gas/vapor response changes when photoinduced processes are active. Similar behavior of the m-SWNTs and MWNTs confirms metallic behavior of the MWNT film due to delocalization of conduction band electrons. The adsorption of electron-accepting O_2 and H_2O on these networks leads to charge carrier depletion in the film surface, increasing the resistance. Under UV irradiation, electrons are photoexcited to conduction bands in the depleted region leaving behind holes that contribute to charge transport.

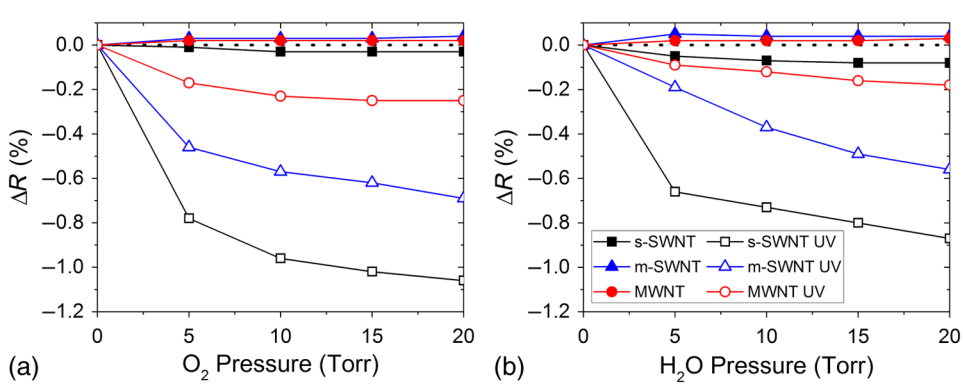

Fig. 5 Resistance response (ΔR) to (a) O_2 and (b) H_2O as a function of gas/vapor pressure in the dark (solid shapes) and under UV irradiation (open shapes). $\Delta R = 0$ is denoted by the dotted horizontal line.

Table 1 O_2 and H_2O detection limits for CNT networks in the dark and under UV irradiation.

CNT network	O_2 detection limit (Torr)	H_2O detection limit (Torr)
s-SWNT	2.48 ± 0.02	0.16 ± 0.03
s-SWNT UV	0.04 ± 0.02	0.05 ± 0.02
m-SWNT	1.09 ± 0.01	1.42 ± 0.02
m-SWNT UV	0.10 ± 0.01	0.14 ± 0.02
MWNT	3.32 ± 0.02	2.48 ± 0.02
MWNT UV	0.13 ± 0.02	0.15 ± 0.03

Consequently, UV irradiation induces p-type behavior in metallic CNTs when electron acceptor adsorbates are present on the film. Since s-SWNTs exhibit p-type behavior, adsorption of acceptor molecules in the dark increases the film conductivity, as seen in Figs. 5(a) and 5(b). Under UV irradiation, electronic photoexcitations result in an increase in valence band holes, further increasing the film conductivity.

The data shown in Figs. 2(c) and 2(d) were extrapolated to estimate the limits of H_2O and O_2 detection for each CNT network in the dark and under UV irradiation, as shown in Table 1. Here, detection limits are defined as minimum gas/vapor pressures required to induce resistance changes of $>0.2\ \Omega$ in the CNT network. For reference, the lowest H_2O detection limit reported here for s-SWNT under UV irradiation corresponds to $0.20 \pm 0.08\%$ relative humidity. The detection limits drop by roughly an order of magnitude or more when CNT networks are exposed to UV irradiation. This suggests that while UV irradiation is important for improving gas/vapor sensor sensitivity, it also decreases the electronic stability of CNT films undergoing sorption by ambient gas/vapor molecules.

4 Conclusions

Interaction between O_2 and H_2O and CNT networks depends on the nature of nanotubes (semiconducting, metallic). In metallic nanotubes, O_2 and H_2O adsorption decreases concentration of mobile charge carriers (electrons), resulting in higher resistance due to formation of ($CNT^{\delta+\cdots}[\text{acceptor}]^{\delta-}$) charge transfer complex. In s-SWNT networks, the effect is opposite due to p-type behavior. UV irradiation reverses the sign of majority charge carriers in metallic nanotubes, and under O_2 and H_2O exposure the network resistance decreases. In s-SWNT networks, the nature of majority charge carriers does not change and adsorption of electron acceptors further increases the conductivity of the network. In s-SWNT, we find evidence of adsorption-induced disruption of CNT adhesion to gold electrodes. We propose that the effect is reversible and stems from perturbation of van der Waals interaction between tubes and preferential adsorption of electron acceptors on gold, which leads to partial delamination of s-SWNTs from the QCM surface. We find that UV irradiation increases the resistive response and decreases the detection limits of s-SWNT, m-SWNT, and MWNT networks by roughly an order of magnitude. This demonstrates the importance of UV irradiation for improving the sensitivity of gas/vapor sensors and elucidates the role of UV irradiation on the environmental stability of CNT-based devices.

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